

SAIPAN, Northern Mariana Islands

WMO: 912320

Lat: 15.117N

Lon: 145.717E

Elev: 66

StdP: 100.54

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 41418

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.0	23.8	17.9	12.9	25.3	18.9	13.8	25.7	12.3	26.1	11.4	26.7	4.4	10	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.8	32.1	26.6	31.6	26.4	31.2	26.3	27.7	30.3	27.4	30.0	27.1	29.8	5.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.9	29.0	26.6	22.3	28.8	26.2	21.8	28.7	89.0	30.8	87.4	30.1	86.2	30.2	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.8	9.1	21.6	33.9	1.3	2.1	20.7	35.4	19.9	36.6	19.2	37.8	18.2	39.3		
(5)			18.8	28.6	1.0	1.2	18.1	29.4	17.5	30.1	16.9	30.8	16.1	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	26.7	26.5	27.0	27.7	28.3	28.7	28.2	28.1	27.9	27.8	27.9	27.5	(6)
(7)		DBStd	1.14	0.98	0.89	0.93	0.85	0.71	0.93	1.10	1.11	1.01	1.00	0.88	0.87	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6459	518	463	526	531	568	561	566	560	537	552	537	541	(10)
(11)		CDD18.3	3417	260	229	267	281	310	311	307	302	287	293	287	283	(11)
(12)		CDH23.3	36139	2309	1964	2463	2942	3514	3719	3535	3389	3149	3172	3102	2882	(12)
(13)		CDH26.7	10152	409	329	521	793	1134	1385	1191	1063	926	888	861	651	(13)
(14)	Wind	WSAvg	5.1	6.0	6.2	6.1	5.7	4.9	4.7	4.1	3.8	3.7	4.5	5.8	6.2	(14)
(15)	Precipitation	PrecAvg	1990	98	86	68	79	80	125	245	326	293	279	191	129	(15)
(16)		PrecMax	2439	207	239	166	255	164	467	505	725	445	476	336	322	(16)
(17)		PrecMin	1186	28	22	9	16	28	20	140	175	142	158	39	43	(17)
(18)		PrecStd	272	57	54	36	50	36	87	84	136	76	92	82	65	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	30.7	31.1	31.6	32.0	33.0	32.9	33.3	32.1	31.4	31.3	31.0	(19)
(20)			MCWB	25.6	25.2	25.3	25.7	26.1	26.9	27.1	27.4	27.1	26.8	26.4	25.8	(20)
(21)		2%	DB	30.1	29.7	30.2	31.0	31.2	32.0	31.7	31.5	31.2	31.0	30.8	30.2	(21)
(22)			MCWB	24.9	24.7	24.9	25.4	25.9	26.4	26.7	26.9	26.7	26.7	26.3	25.6	(22)
(23)		5%	DB	29.3	29.1	29.7	30.4	31.0	31.4	31.1	30.9	30.8	30.5	30.2	29.9	(23)
(24)			MCWB	24.7	24.5	24.7	25.3	25.8	26.3	26.6	26.7	26.6	26.6	26.2	25.4	(24)
(25)		10%	DB	28.9	28.7	29.1	29.9	30.5	31.0	30.6	30.3	30.2	30.0	29.9	29.3	(25)
(26)			MCWB	24.5	24.3	24.4	25.2	25.7	26.2	26.5	26.5	26.5	26.4	26.1	25.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.6	26.3	27.0	27.4	28.2	28.0	28.0	27.9	27.7	27.2	(27)	
(28)			MCD B	29.4	28.7	29.5	29.7	30.3	31.0	30.4	31.0	30.6	30.0	29.9	29.2	(28)
(29)		2%	WB	26.0	26.0	25.7	26.4	26.8	27.1	27.6	27.5	27.4	27.2	27.1	26.6	(29)
(30)			MCD B	28.5	28.3	28.9	29.3	29.9	30.6	30.0	30.3	30.0	29.5	29.4	28.7	(30)
(31)		5%	WB	25.6	25.4	25.3	26.0	26.4	26.8	27.2	27.1	27.1	27.0	26.7	26.1	(31)
(32)			MCD B	28.0	27.8	28.3	29.1	29.5	30.1	29.8	29.8	29.6	29.3	29.0	28.4	(32)
(33)		10%	WB	25.2	25.0	24.9	25.6	26.1	26.5	26.8	26.8	26.7	26.7	26.4	25.8	(33)
(34)			MCD B	27.6	27.4	27.9	28.7	29.2	29.8	29.5	29.3	29.1	28.9	28.7	28.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	4.5	4.7	4.9	4.7	4.8	4.6	4.6	4.7	4.5	4.3	4.1	(35)
(36)			MCDBR	5.1	5.0	5.2	5.4	5.1	5.3	5.1	5.2	5.3	4.9	4.6	4.6	(36)
(37)			MCWBR	2.5	2.6	2.4	2.4	2.2	2.3	2.4	2.6	2.6	2.5	2.4	2.4	(37)
(38)		5% WB	MCD BR	4.7	4.6	4.7	4.9	4.7	5.0	4.7	4.8	4.9	4.5	4.3	4.0	(38)
(39)			MCWBR	2.6	2.7	2.5	2.5	2.4	2.4	2.5	2.6	2.6	2.6	2.6	2.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.407	0.410	0.406	0.410	0.410	0.411	0.406	0.414	0.405	0.428	0.432	0.418	(40)	
(41)		taud	2.485	2.470	2.495	2.515	2.537	2.537	2.566	2.538	2.570	2.489	2.423	2.449	(41)	
(42)		Ebn at Noon	887	899	909	898	885	875	881	883	897	872	854	864	(42)	
(43)		Edh at Noon	106	112	112	109	105	104	102	106	102	108	112	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.84	5.54	6.36	6.88	6.91	6.67	5.87	5.25	5.27	5.08	5.11	4.81	(44)	
(45)		RadStd	0.27	0.28	0.25	0.30	0.24	0.47	0.44	0.49	0.34	0.49	0.27	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+109
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page